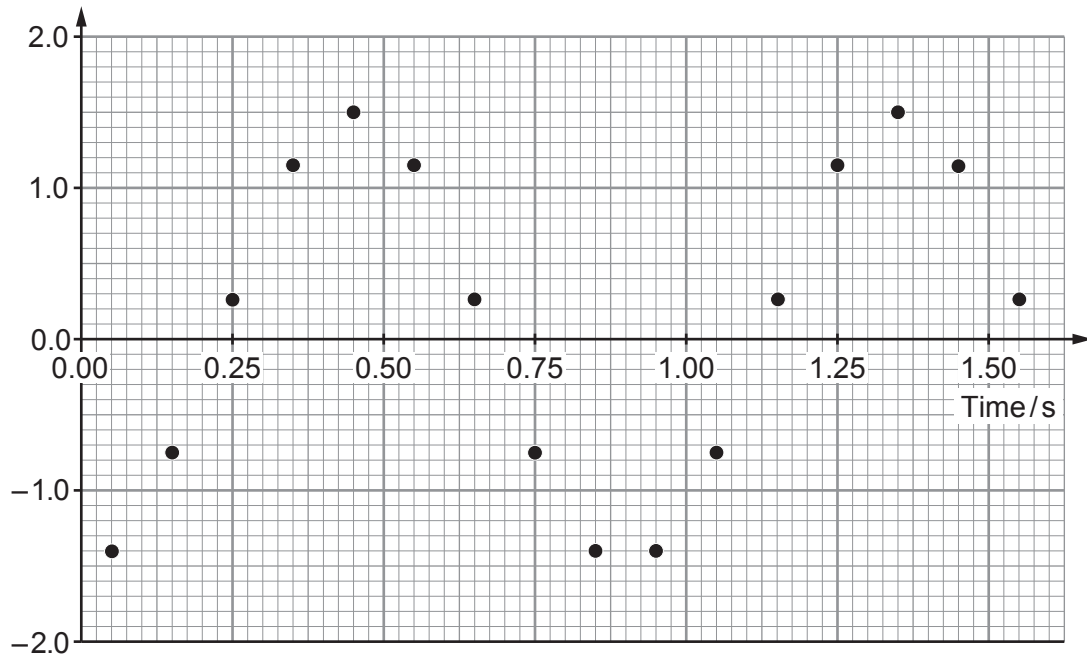


4. A 0.030 kg mass is suspended from a light, vertical spring. A student sets the mass to oscillate vertically. The position of the oscillating mass is tracked using a datalogger. Measurements, taken over a time interval of approximately 1.50 s, are shown on the graph below. The amplitude during this interval was 1.5 cm.

Displacement/cm



- (a) **Draw on the grid** above the curve of best fit. [2]

- (b) Explain what is meant by the 'period of the oscillation'. Use the graph to support your answer. [2]

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- (c) **Complete** the expression for the displacement, y , of the mass. [2]

$$y = 1.5 \cos \left(\left(\frac{2\pi}{\dots\dots\dots} \right) t \dots\dots\dots \right) \text{ cm}$$



- (d) (i) By drawing a tangent to the displacement-time curve, show that the maximum velocity of the oscillating mass is approximately 10 cm s^{-1} . [3]

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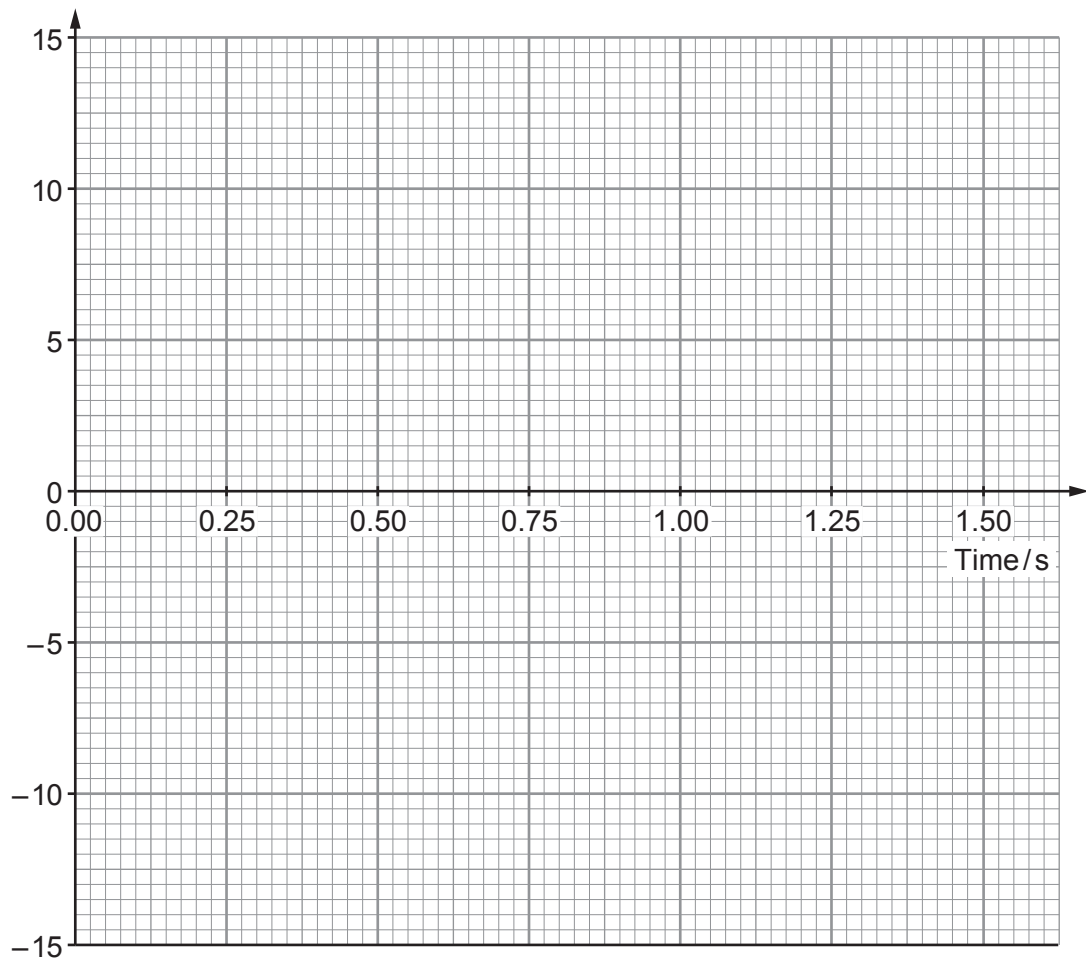
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- (ii) **Sketch** the velocity-time graph for the oscillating mass on the grid below. [3]

Velocity/ cm s^{-1}



(e) Catrin suggests that the 0.030 kg mass is oscillating with simple harmonic motion.

(i) State what is meant by 'simple harmonic motion'. [2]

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(ii) Calculate the value of the spring constant. [3]

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- [6]

